

Lecture 3

September 1

Recap from before

- Early thoughts: observations should not be combined
- Medium thoughts: some observations can be combined (if they are essentially the same)
- Medium + thoughts:

Learning objectives

- How Laplace improved on Mayer's ad hoc combination of observations, and what their methods reveal about error, random error, and researcher judgment.
- How Boscovich used latitude and arc-length measurements to study Earth's shape and introduced a principled best-fit rule by balancing and minimizing corrections.
- How Laplace refined Boscovich's method through weighting, and how both approaches connect to median regression, measurement variability, and modern likelihood ideas.

Error

- Systematic
(the instrument is)
wrong
- Accumulating

Random Error

- If you add them up
they cancel out.
(can be positive or negative)
- Can adjust for them

Mayer knew about some measurement error but
ignored them...

Laplace's Rescue operation

Recall: Euler gave up on his many equations
and so maybe Jupiter will crash into the sun

— Combine 24 equations 4 times

(1) sum all 24 equations (maximize the constant term)

(2) sum the first 12 and subtract the sum of
the next 12 (zero out the constant term)

(3)

(4) combine so that
3rd & 4th sum are small/big

Mayer

vs

Laplace

- ① Only combine disjoint observations
- ② No general principle stated.
- ③ First group have large & coefficients
Second group has large negative & coefficients

Maybe principles

- ① Observations can be used multiple times
- ② Mary Somerville describes the principles implied by Laplace

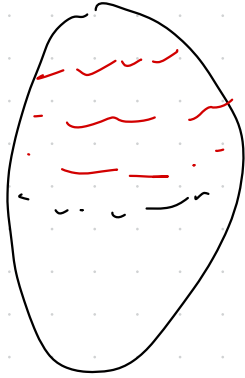
Table 1.3. Laplace's Saturn data.

Eq. no.	Year (i)	$-a_i$	b_i	c_i	d_i	Laplace residual	Halley residual	L.S. residual
1	1591	1'11.9"	-158.0	0.22041	-0.97541	+1'33"	-0'54"	+1'36"
2	1598	3'32.7"	-151.78	0.99974	-0.02278	-0.07	+0.37	+0.05
3	1660	5'12.0"	-89.67	0.79735	0.60352	-1.36	+2.58	-1.21
4	1664	3'56.7"	-85.54	0.04241	0.99910	-0.35	+3.20	-0.29
5	1667	3'31.7"	-82.45	-0.57924	0.81516	-0.21	+3.50	-0.33
6	1672	3'32.8"	-77.28	-0.98890	-0.14858	-0.58	+3.25	-1.06
7	1679	3'9.9"	-70.01	0.12591	-0.99204	-0.14	-1.57	-0.08
8	1687	4'49.2"	-62.79	0.99476	0.10222	-1.09	-4.54	-0.52
9	1690	3'26.8"	-59.66	0.72246	0.69141	+0.25	-7.59	+0.29
10	1694	2'4.9"	-55.52	-0.07303	0.99733	+1.29	-9.00	+1.23
11	1697	2'37.4"	-52.43	-0.66945	0.74285	+0.25	-9.35	+0.22
12	1701	2'41.2"	-48.29	-0.99902	-0.04435	+0.01	-8.00	-0.07
13	1731	3'31.4"	-18.27	-0.98712	-0.15998	-0.47	-4.50	-0.53
14	1738	4'9.5"	-11.01	0.13759	-0.99049	-1.02	-7.49	-0.56
15	1746	4'58.3"	-3.75	0.99348	0.11401	-1.07	-4.21	-0.50
16	1749	4'3.8"	-0.65	0.71410	0.70004	-0.12	-8.38	+0.03
17	1753	1'58.2"	3.48	-0.08518	0.99637	+1.54	-13.39	+1.41
18	1756	1'35.2"	6.58	-0.67859	0.73452	+1.37	-17.27	+1.35
19	1760	3'14.0"	10.72	-0.99838	-0.05691	-0.23	-22.17	-0.29
20	1767	1'40.2"	17.98	0.03403	-0.99942	+1.29	-13.12	+1.34
21	1775	3'46.0"	25.23	0.99994	0.01065	+0.19	+2.12	+0.26
22	1778	4'32.9"	28.33	0.78255	0.62559	-0.34	+1.21	-0.19
23	1782	4'4.4"	32.46	0.01794	0.99984	-0.23	-5.18	-0.15
24	1785	4'17.6"	35.56	-0.59930	0.80053	-0.56	-12.07	-0.57

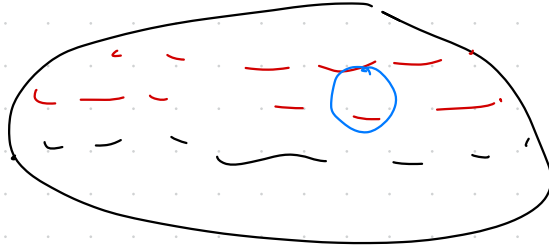
Source: Laplace (1788).

Note: Residuals are fitted values minus observed values.

Another problem: Shape of the earth



prolate



oblate

Learning A just
Europe has
limited
Latitudes

- Confirms / Denies Newton's ideas

- How to measure: measure arc length
at different latitudes.

What is Latitude

- Degrees from center of the earth to the equator
Geocentric Latitude
- Gemini gave the above definition
without qualification

What is Latitude

- The projection of a
ray from the
edge of the earth
to the equatorial plane
Geodetic Latitude

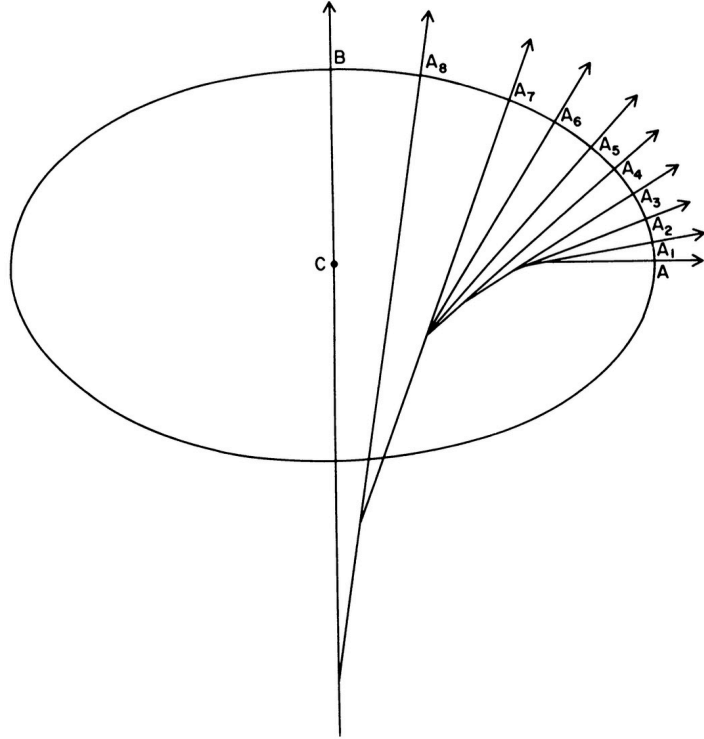


Figure 1.3. A side view of an exaggeratedly oblate earth, illustrating the lengthening of degrees of arc toward the pole. The meridian quadrant AB is broken into nine segments, each of 10° latitude. (Based upon Berry, 1898, p. 277.)

Arc length and latitude An elliptical integral -

an
approximation

Let a be an arc length of a 1 degree of latitude
centered at latitude θ then

$$a = z + y \sin^2 \theta$$

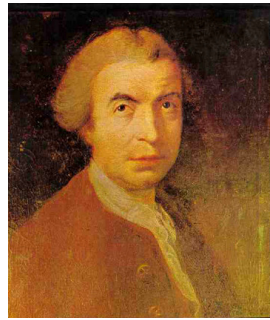
z is length of a 1 degree arc at
equator

y is the excess of a degree at
the north pole over one at the
equator

Boscovich, the man

1711-1787, born in Dubrovnik, Ragusa

Diplomat and ...



- Sent to London during 7 years war to persuade the British that Ragusa was not helping the French.
- Used trip to promote himself - elected member of Royal Society.
- Wrote a 5000 line poem about Astronomy that is described as "uninstructive to an astronomer and unintelligible to anyone else" (Delambre)
- Helped fix St. Peter's dome by suggesting metal chains be put around it!
- Helped start an observatory in Milan, then was kicked out
- Argued to allow Copernicus's books to be read.

Boscovich's 5 observations

Table 1.4. Boscovich's data on meridian arcs.

Location	Latitude (θ)	Arc length (toises)	Boscovich's $\sin^2 \theta \times 10^4$
(1) Quito	$0^\circ 0'$	56,751	0
(2) Cape of Good Hope	$33^\circ 18'$	57,037	2,987
(3) Rome	$42^\circ 59'$	56,979	4,648
(4) Paris	$49^\circ 23'$	57,074	5,762
(5) Lapland	$66^\circ 19'$	57,422	8,386

Source: Boscovich and Maire (1755, p. 500). Reprinted in Boscovich and Maire (1770, p. 482).

Note: Arc lengths are given as toises per degree measured, where 1 toise $\cong$ 6.39 feet. The value for $\sin^2 \theta \times 10^4$ for the Cape of Good Hope is erroneous and is evidently based on $33^\circ 8'$. The correct figure would be 3,014.

- Gets 10 calculations, gets stuck.
- Early p-hacking

The method

The mean we will take will not be a simple arithmetic mean, rather it will be one tied by a certain law to the rules of fortuitous combination and the calculus of probabilities. We are faced here with a problem I have discussed toward the end of a Dissertation inserted in the proceedings of the Institute of Bologna, volume 4 [that is, Boscovich's 1757 summary], where I contented myself with giving the result of its solution. Here is the problem: *Being given a certain number of degrees, find the correction that must be made to each of them, supposing these three conditions are complied with: the first, that their differences shall be proportional to the differences between the versed sines of twice their latitudes; the second, that the sum of the positive corrections shall be equal to the sum of the negative ones; the third, that the sum of all the corrections, positive as well as negative, shall be the least possible, for the case where the first two conditions will be fulfilled.* The first condition is called for by the law of equilibrium, which requires an elliptical shape; the second, from the fact that deviations of a pendulum, or errors by observers, that augment or diminish degrees have the same degree of probability; the third is necessary in order to approximate the observations as closely as possible, for, as we have observed above, it is clearly very probable that the deviations are quite small, because the scrupulous exactitude of the observers would not permit any suspicion of large errors in their observations. (Boscovich, 1760, as translated from the French of Boscovich and Maire, 1770, p. 501)

- ④ Some "shape" for the problem
- ② errors cancel out

The math

$$fa_i = z + y \sin^2 \theta_i - a_i$$

minimize $\sum_i |fa_i|$ such that $\sum_i fa_i = 0$

$|x_i| = \text{positive version of } x_i$
 $\sum_i |x_i| = |x_1| + |x_2| + \dots + |x_n|$

In the future $\rightarrow$ call this
median regression

$$\min_{z, y} \sum_i |z + y \sin^2 \theta_i - a_i|$$

What did Boscorich do?

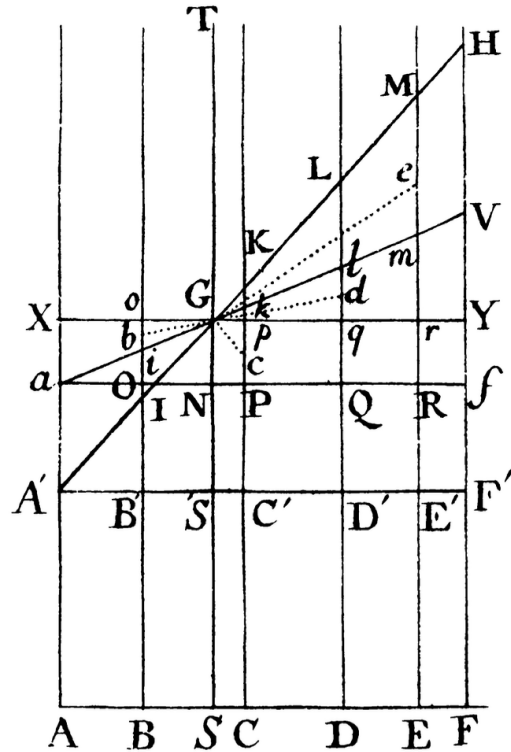


Figure 1.4. Boscovich's algorithm. The horizontal axis AF gives $\sin^2 \theta$, where θ is the latitude of the midpoint of the arc; the vertical axis AX gives the arc length in toises per degree. The five arcs are indicated by a, b, c, d , and e ; G is the center of gravity. (From Boscovich and Maire, 1770, plate I.)

Laplace, again.

First, that the sum of the errors committed in the measures of the whole arcs, ought to be zero. Second, that the sum of all these errors, taken positively, ought to be a *minimum*. By considering, in this manner, the whole arcs, instead of the degrees which have been deduced from them, we shall give to each of these degrees so much more influence, in the computation of the ellipticity of the earth, as the corresponding arc is of greater extent, which ought to be the case. (Laplace, 1799–1805, vol. 2, p. 134; Bowditch, 1832, pp. 434–437)

Sources of Error

Write $a_i = y \sin^2 \theta_i + \epsilon_i$

ϵ_i is our error.

